

Cloud Computing Laboratory Manual

Department: Information Technology

Course: Cloud Computing Laboratory

Academic Year: 2025-26

Objective:

This laboratory manual is designed for faculty to guide students in performing cloud computing experiments. It provides theoretical background, step-by-step procedures, expected outputs, and evaluation guidelines for each experiment.

Experiment 1: Installation of Google App Engine and Hello World Application

Aim:

To install Google App Engine environment and deploy a simple Hello World web application.

Theory:

Google App Engine is a Platform as a Service (PaaS) that allows developers to build and host web applications in Google-managed data centers.

It provides automatic scaling, load balancing, and application health monitoring. Developers can deploy applications written in Python, Java,

Node.js, Go, and other supported languages.

Prerequisites:

- Basic knowledge of web programming
- Python installation
- Google account
- Internet connection

Step-by-Step Procedure:

1. Install Google Cloud SDK from the official website.
2. Open terminal or command prompt.
3. Initialize SDK using the command: `gcloud init`
4. Create a new project in Google Cloud Console.
5. Create a folder named hello-app.
6. Inside the folder create file `main.py`.
7. Write Python Flask code to display Hello World.
8. Create configuration file `app.yaml`.
9. Deploy application using command `gcloud app deploy`.
10. Launch application using `gcloud app browse`.

Sample Code:

```
from flask import Flask  
app = Flask(__name__)
```

```
@app.route('/')  
def hello():
```

```
    return "Hello World from Google App Engine"
```

Expected Output:

Web browser displays Hello World message hosted on cloud platform.

Learning Outcome:

Students understand deployment of web applications on cloud infrastructure.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 2: Launch Web Application using Google App Engine Launcher

Aim:

To run and test a web application locally using the Google App Engine Launcher.

Theory:

Local testing allows developers to debug applications before deploying them to the cloud environment.

The App Engine launcher provides tools to run applications on a local development server.

Procedure:

1. Install Google App Engine Launcher.
2. Open the launcher application.
3. Click Add Existing Application.
4. Select the project directory created earlier.
5. Click Run.
6. Wait for the development server to start.
7. Open browser and navigate to `http://localhost:8080`

Expected Output:

The Hello World application runs on the local server.

Learning Outcome:

Students learn how to test cloud applications locally before deployment.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 3: Cloud Simulation using CloudSim

Aim:

To simulate a cloud computing environment using CloudSim toolkit.

Theory:

CloudSim is a framework developed for modeling and simulation of cloud computing environments.

It enables researchers and developers to test scheduling algorithms, resource allocation strategies,

and virtual machine provisioning without using real cloud infrastructure.

Procedure:

1. Install Java Development Kit.
2. Install Eclipse IDE.
3. Download CloudSim library.
4. Create new Java project in Eclipse.
5. Add CloudSim jar files to project libraries.
6. Write simulation code to create datacenter, hosts, virtual machines, and cloudlets.
7. Implement custom scheduling algorithm.
8. Run simulation and observe results.

Expected Output:

Simulation results showing task allocation and execution time.

Learning Outcome:

Students understand scheduling algorithms and resource management in cloud computing.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 4: Transfer Files Between Virtual Machines

Aim:

To transfer files securely between two virtual machines.

Theory:

In distributed cloud environments, virtual machines often need to exchange data.

Secure Copy Protocol (SCP) is commonly used for encrypted file transfer between systems.

Procedure:

1. Start two virtual machines.
2. Identify IP addresses using command `ifconfig` or `ipconfig`.
3. Ensure both machines are connected to network.
4. Use SCP command for file transfer.

Example Command:

```
scp sample.txt user@192.168.1.20:/home/user/
```

Expected Output:

File appears in the destination machine directory.

Learning Outcome:

Students learn secure file transfer methods in virtualized environments.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 5: Launch Virtual Machine using TryStack (OpenStack)

Aim:

To launch and manage virtual machines using OpenStack cloud platform.

Theory:

OpenStack is an open-source cloud computing platform that controls large pools of compute, storage,

and networking resources. TryStack provides a free demonstration environment for learning OpenStack.

Procedure:

1. Create TryStack account.
2. Login to OpenStack Horizon Dashboard.
3. Navigate to Compute → Instances.
4. Click Launch Instance.
5. Select operating system image (Ubuntu).
6. Choose instance flavor.
7. Configure network settings.
8. Launch instance.

Expected Output:

Virtual machine instance created and running in dashboard.

Learning Outcome:

Students understand infrastructure provisioning in cloud platforms.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 6: Deploy Web Application in PaaS Environment

Aim:

To design and deploy a simple web application using a Platform as a Service environment.

Theory:

PaaS platforms provide development tools and runtime environments for building applications without

managing underlying hardware or operating systems.

Procedure:

1. Create simple HTML web page.
2. Create backend script if required.
3. Configure deployment files.
4. Deploy application using Google Cloud commands.
5. Access application using browser URL.

Expected Output:

Web application accessible through internet.

Learning Outcome:

Students learn deployment lifecycle of cloud applications.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 7: Develop Custom Application using Salesforce Cloud

Aim:

To design and develop a simple business application using Salesforce cloud platform.

Theory:

Salesforce is a leading cloud-based Customer Relationship Management platform that allows development

of enterprise applications using its cloud ecosystem.

Procedure:

1. Create Salesforce developer account.
2. Login to Salesforce dashboard.
3. Create custom object.
4. Add fields such as Name, Email, Phone.
5. Design application layout.
6. Insert and manage records.

Expected Output:

Application interface allows storing and retrieving records.

Learning Outcome:

Students understand cloud-based enterprise application development.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?

Experiment 8: Firebase Authentication and Google Cloud Datastore

Aim:

To authenticate users using Firebase Authentication and store user data in cloud datastore.

Theory:

Firebase Authentication provides secure user authentication services. It supports email/password,

Google login, and other identity providers.

Procedure:

1. Create Firebase project.
2. Enable Email/Password authentication.
3. Create login form in HTML.
4. Integrate Firebase authentication script.
5. Verify user credentials.
6. Store user data in Google Cloud Datastore.

Expected Output:

User login is verified and credentials are stored securely in cloud database.

Learning Outcome:

Students learn cloud-based authentication and database integration.

Conclusion

Students successfully completed the experiment and understood the concept demonstrated.

Viva Questions

1. Explain the concept related to this experiment.
2. What are the advantages of using cloud computing?
3. What are the limitations of the implemented approach?